

ATTACHMENT 1

Wildfire Imagery Pilot–Statement of Work

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I. INTRODUCTION

The National Oceanic and Atmospheric Administration’s (NOAA) mission is to understand and predict changes in climate, weather, oceans, and coasts, to share that knowledge and information with others as the Nation’s authoritative environmental intelligence agency. NOAA is committed to improving public predictions to protect life and property.

Wildfires pose a significant threat to communities across the United States, and NOAA's observations are crucial for both fire mitigation and response efforts. In particular, early fire detection and monitoring of fire intensity and progression are critical for first response and wildfire incident management, both of which the National Weather Service supports. NOAA seeks to augment its remote sensing fire detection and monitoring capabilities with commercial space-based multispectral capabilities. These commercial observations will aid in modernizing fire warnings and synthesizing data from multiple sources to support time-critical decision-making.

The NESDIS Commercial Data Program (CDP) pursues demonstration projects on behalf of NOAA and assesses the viability of assimilating commercially provided satellite data and products to diversify NOAA’s portfolio of data collection capabilities. Through the NESDIS CDP, NOAA intends

to conduct a commercial data Pilot Study that may help NOAA meet its terrestrial environment mission objectives. For this Pilot Study, NOAA will use Pilot Study data to assess the quality and impact of available commercial multispectral imagery, including infrared capabilities, and to investigate the utility of radiometric data developed by commercial vendors to aid in wildfire detection, monitoring, and mapping. NOAA will assess thermal-based fire detection performance compared to NOAA's Advanced Baseline Imager (ABI) and Visible Infrared Imaging Radiometer Suite (VIIRS) instruments. NOAA will use these observations to generate fire detection and characterization intelligence and to assess their value within an integrated data and application framework. NOAA will focus these efforts on use cases related to protecting life and property.

II. DATA SPECIFICATION REQUIREMENTS

The Contractor for this Pilot Study shall supply multi-spectral observation data that, preferably, includes visible, near infrared, short-wave infrared, mid-wave infrared, and long-wave infrared channels collected on-orbit from one or more satellites. The observation data will primarily be used to evaluate commercial multispectral imagery for wildfire detection, characterization, monitoring, and mapping.

1. The requirements set forth below constitute the specific operational measurement requirements for the Pilot Study. Requirements are categorized as *threshold* values, representing minimum acceptable levels, and *objective* values, representing desired levels of performance with increased utility. Threshold and objective requirements specified in this document are not necessarily indicative of requirements that may be developed for future contracts or solicitations.
2. The threshold requirements define the minimum acceptable performance levels; failure to meet them may result in no contract award.
3. For the objective requirements, it is understood that contractors may not be able to meet all objective requirements. Contractors proposing to deliver data that do not meet one or more objective requirements during the data delivery period shall clearly identify:
 - a. the requirement(s) not met,
 - b. the reason(s) for not meeting the requirement(s), and
 - c. the expected impact on data quality, operational usability, and evaluation outcomes.
4. Due to the nature of this Pilot Study, it is understood that contractors may not be able to meet all requirements presented. Contractors proposing to deliver data that does not meet a requirement during the Pilot Study data delivery period should indicate the nature of the discrepancy, the reason for not meeting the requirement, and the expected impact.

II.A. DEFINITIONS

1. NOAA Data Processing Level Definitions include:

Table 1. NOAA Data Processing Level

Data Level	Description	Multispectral Imagery Examples
Level 0	Reconstructed, unprocessed instrument and payload data at full resolution, with any and all communications artifacts (e.g., synchronization frames, communications headers, duplicate data) removed.	Raw detector counts from each channel, time stamps for each scan line, scan geometry and sensor view angles, instrument telemetry.
Level 1A (L1A)	Level-1A (L1A) data are reconstructed and time-ordered instrument measurements that have been de-packetized and quality-checked but remain uncalibrated.	Uncalibrated digital counts from each channel with time, scan angle, and instrument telemetry.
Level 1B (L1B)	Level-1B (L1B) data contain radiometrically calibrated and geolocated radiance measurements for each spectral band. Processing includes: conversion from raw counts to physical radiance units; application of calibration coefficients; assignment of Earth-location coordinates (geolocation) and quality flags and metadata. L1B data are the primary input for generating geophysical products.	Calibrated, geolocated radiances or reflectances for each spectral band and spatial pixel.
Level 1C (L1C)	Resampled and geolocated brightness temperature data projected onto a standard Earth grid, enabling multi-sensor analysis and model ingestion.	Brightness temperatures gridded to a fixed Earth coordinate system (e.g., EASE-Grid).
Level 2 (L2)	Level-2 (L2) data are derived geophysical products — such as cloud properties, sea-surface temperature, aerosol optical depth, fire detections and fire radiative power, and vegetation indices — computed from Level-1B radiances and ready for scientific and operational use.	Cloud properties, sea-surface temperature, aerosol optical depth, fire detections and fire radiative power

2. **Metadata** is structured, descriptive information that documents the characteristics of a dataset. They provide essential context about the data's content, origin, purpose, quality, format, spatial and temporal coverage, processing history, and usage constraints. Metadata enables effective data discovery, interpretation, integration, management, and reuse by both humans and machines. High-quality metadata are critical for ensuring transparency, traceability, and interoperability within NOAA's data enterprise and across the broader scientific community.

For additional guidance on metadata standards and practices, see NOAA’s National Centers for Environmental Information (NCEI):

<https://www.ncei.noaa.gov/resources/metadata/introduction-to-metadata-at-ncei>

3. **Latency** of a data product is defined as the difference between the time the instrument acquires data in orbit and the time the data are received into the NESDIS Common Cloud Framework (NCCF) or alternate NESDIS cloud environment.
4. **Satellite data availability** is the percentage of satellite-generated data that, during a given time period, is accessible, usable, and delivered as expected to users. It reflects the overall capability of the satellite system—including onboard sensors, communication links, ground stations, and processing infrastructure—to provide data continuously and without interruption.

II.B. OBSERVATION CAPABILITY AND REQUIREMENTS

1. The Contractor shall supply multispectral observation data. The requirements set forth below constitute the measurement requirements.
2. Spectral acquisitions must remain within the Earth exploration satellite service (EESS) allocations, as specified in the [FCC Online Table of Frequency Allocations](#), to avoid compromising system performance or violating regulatory constraints.
3. Table 2, summarized below, lists the spectral guidance and acceptable capabilities for multispectral radiance observations for Fire Imagery. The Contractor shall provide radiance observations that, at a minimum, include at least one channel in each of the specified spectral ranges. Additional spectral observations, if available, shall also be provided.

Table 2. Spectral Guidance for Multispectral Radiance Observations for Fire Detection Imagery

Spectral Band	Spectral Channel Ranges (μm)	Primary Use
Visible / Near Infrared	0.6 - 0.9	False Alarm Screening
Shortwave / Midwave Infrared	3.5 - 4.2	Active Fire Detection and Characterization
Longwave Infrared	8.0 - 12.5	Active Fire Detection and False Alarm Screening

5. Table 3, summarized below, presents the observational requirements for multispectral imagery for fire imagery detection. The Contractor shall adhere to these requirements and meet or exceed the stated thresholds in Table 3.

The contractor shall deliver at least 80% of the constellation-level data (full-swath data) over a 3-9 month period during peak wildfire season (June - November), when any part of the full satellite swath intersects a predefined region of interest, with a primary focus on the western United States. The contractor may provide two pricing options: one for a region of interest that is 1900 km in the north/south direction and 1000 km in the east/west direction, and another for a 950 km (N/S) x 500 km (E/W) region. The location of the region of interest will be based on wildfire activity during the 3-9 month pilot period.

Table 3. Multispectral Radiance Data Requirements for Fire Detection Imagery

Performance Level	Threshold Requirement
Horizontal Spatial Resolution (at nadir)	Threshold: 200 m, Objective: 50 m
Geolocation Accuracy	Threshold: 100% of Horizontal spatial resolution, Objective: 50% of Horizontal spatial resolution
L1B product Latency	Threshold: 3 hours, Objective: 60 minutes

III. DELIVERY REQUIREMENTS

Contractors shall supply data that meets the following requirements:

1. Provide Level 1 data products consistent with or improving upon the spatial and radiometric data characteristics of NOAA's [Visible Infrared Imaging Radiometer Suite \(VIIRS\)](#) sensor. VIIRS serves as references for spatial resolution, radiometric performance, and geolocation accuracy.

As defined in Section II.A, data products shall include, at a minimum:

- a. Level 1B: Calibrated and geolocated radiances in $\text{Wm}^{-2}\text{sr}^{-1}\mu\text{m}^{-1}$ units along with Lookup Tables (LUTs) or band-specific information to convert them to reflectance (unitless) and brightness temperatures (Kelvin). These radiometrically calibrated measurements shall be traceable to onboard blackbody references or other high-fidelity validation references and reported at the sensor's native resolution. Level 1B products shall be formatted as NetCDF, HDF5, or geotiff files.
- b. Level 1C: Level 1B data resampled onto an orthomosaic using a fixed Earth grid. Level 1C products shall be formatted as NetCDF, HDF5, or geotiff files.

2. The Contractor shall ensure continuous and contiguous data coverage over geographic regions and periods of interest, and deliver all collected data without interruptions, providing a complete record of observations in near real-time.
 - a. At a minimum, the Contractor must have the capability to provide contiguous data, without temporal or spatial gaps, along-track for one satellite. Cross-track coverage shall be sufficient to meet the spatial and temporal coverage requirements as specified in this SOW.
3. To ensure consistent delivery of supplementary commercial multispectral radiometric data, overall constellation-level data availability shall serve as the primary performance metric. The Contractor shall maintain at least 80% operational availability. This availability accounts for routine maintenance, calibration activities, and orbital dynamics, and ensures that the spacecraft are capable of supporting NOAA's data acquisition needs. This constellation-level approach ensures sufficient data continuity to support NOAA mission needs while allowing flexibility in individual satellite performance.
4. The product latency of multispectral radiometric observations delivered by the Contractor, when averaged over any continuous thirty-day period, shall meet or exceed the required latency as specified in Table 3.
5. The Contractor shall ensure cross-platform data consistency across their constellation of commercial multispectral instruments. All instruments shall support "mix-and-match" interoperability, enabling seamless integration of radiance data regardless of satellite platform.
 - a. All instruments must have consistent calibration algorithms and gain models post-launch, with updates synchronized across the constellation.
 - b. Data products must be traceable to recognized reference standards, with VIIRS designated as the primary external reference instrument.
 - c. Timestamps associated with all mission data—including metadata, telemetry, and science observations—must maintain consistency across satellites to within ± 1 second to ensure compatibility in time-sensitive processing workflows.
 - d. The Contractor shall maintain all technical documentation, including calibration history and consistency reports, and ensure it is accessible to the Government in accordance with applicable data access provisions.
6. In the event of data anomalies or as requested by the Government, the Contractor shall provide Level 0 data to support diagnostic analysis and root cause investigation and/or to

perform independent calibration/validation by the Government or its designated representatives.

7. All satellite or instrument configuration changes that may impact the quality, calibration, or continuity of science data shall be documented in a change log. Notifications of such changes shall be provided to the designated Government point of contact within 24 hours.
8. The Contractor shall provide all necessary technical documentation to support data understanding, usage, and validation. At a minimum, this shall include:
 - a. Algorithm Theoretical Basis Documents (ATBDs) – Describing the scientific rationale, processing methodology, and assumptions underlying data generation algorithms.
 - b. Documentation and data files to support characterization, quality control, and calibration/validation activities, including but not limited to:
 - i. Spectral response functions.
 - ii. Calibration coefficients to convert pixel-based digital values into radiance, and the reflectance and brightness temperature equivalents.
 - iii. Pixel-based geolocation information, including terrain correction.
 - c. Pre-launch and on-orbit calibration reports.
 - d. Documentation of quality control flags and calibration/validation indicators.
 - e. User guides and detailed data specification documents.
9. Data delivery shall follow the following processes:
 - a. The data delivery process along with file naming conventions, data formats, and metadata formats shall follow standard [NESDIS methods and protocols](#). The Contractor shall deliver all files for secure ingest to the NESDIS Common Cloud Framework (NCCF) in accordance with the NCCF Interface Control Document (ICD) or alternative secure ingest methods such as the NESDIS Innovation Hub cloud services. The Contractor shall, upon receipt of alert notifications, resolve and (as needed) redeliver data files when data latency requirements can still be met. Redelivered file names shall include a date or identifier unique to that delivery.
 - b. The data descriptor files shall conform to the [International Organization for Standardization](#) (ISO) 19115-1:2014 and applicable amendments (content) and ISO 19139-1:2019 and applicable amendments (XML schema) standards.

- c. Data shall be provided only from instruments on satellites with an assigned World Meteorological Organization (WMO) satellite identifier or a unique, internationally recognized identifier such as the COSPAR ID.
- d. The data delivery process shall follow NOAA security requirements (see Section VI. SECURITY REQUIREMENTS).

IIIA. OPTIONAL DELIVERY REQUIREMENTS

Contractors should consider supplying optional data that meets the following requirements:

1. Provide Level 2 (if available) data products consistent with or improving upon the spatial and radiometric data characteristics of NOAA's [Visible Infrared Imaging Radiometer Suite \(VIIRS\)](#) mission. VIIRS serves as references for spatial resolution, radiometric performance, and geolocation accuracy.

As defined in Section II.A, data products shall include:

- c. Level 2 (if available): Sparse data array(s) containing pixel-based active fire detection information, including essential attributes such as date and time of observation in UTC, center latitude/longitude coordinates, satellite and solar zenith and azimuth angles, effective pixel area, and fire radiative power (FRP) of fire-affected pixels. Optimally provide complementary 2D fire masks describing primary pixel classes (e.g., active fire, clear land, cloud, water), preferably in NetCDF, HDF5, or geotiff format.

IV. CONTRACTING OFFICER'S REPRESENTATIVE

In accordance with CAR Clause 1352.201-72, a Contracting Officer's Representative (COR) will be designated at award. The COR may provide technical direction and has the authority to accept delivery. The COR is not authorized to make any commitments or otherwise obligate the Government or authorize any changes which affect the contract price, terms, or conditions of the award. Any Contractor request for changes shall be referred to the Contracting Officer directly or through the COR. No such changes shall be made without the express written prior authorization of the Contracting Officer.

A Task Manager (TM) will be designated at award. The TM manages the contract cost, schedule, performance and risks. The TM does not have delegated authority to represent the Contracting Officer or bind the Government. The TM cannot change, waive, add, or request any additional specifications or performance.

V. PERFORMANCE SCHEDULE

The Pilot study will be conducted with the following schedule:

Post-Award:

The Contractor shall participate in a post-award kick-off meeting, no later than 2 weeks after award. Additionally, NOAA/NESDIS CDP may require the Contractor to deliver 48-72 hours of on-orbit data to facilitate Government preparation of data ingestion and conversion.

Phase 1: Preparation - 1 month

In this initial phase, the Contractor shall supply the necessary technical documentation which may include format specifications, calibration data, and any other information that will be needed to ingest and process the data. Delivery mechanisms and data handling processes will be communicated to the Contractor. The Contractor will provide engineering support as needed to the Government to enable the acquisition and processing of the supplied datasets. A delivery of 24-hours of sample data may be required during this phase and used to test and adjust the delivery and data flow process as needed.

Phase 2: Data Delivery - 6 months

In this phase, the Contractor shall supply a series of data deliveries for a period of 6 months meeting the requirements specified in this document (see Sections II, III, IV, V, VIII). All proposals must include detailed cost and performance breakdowns for each monthly increment within this range to facilitate a scalable evaluation of the service requirements. The start and end date/time of this data delivery period will be coordinated with the CO and COR.

Phase 3: Evaluation - 3 months

In this final phase, the Contractor will be available for limited reachback engineering/technical support as needed for problem resolution and technical assistance, as the Government evaluates the supplied dataset.

VI. SECURITY REQUIREMENTS

The Contractor shall comply with CAR 1352.239-72, section (i). The Contractor may request COR approval of alternate security accreditation methodologies for compliance with section (i) including: information security assessment methodology promulgated by the U.S Federal Government (specifically, National Institute of Standards and Technology (NIST) Special Publication (SP) 800-37, NIST SP 800- 53, and NIST SP 800-171), or internationally recognized private industry information technology (IT) security frameworks (such as International

Standards Organization (ISO)/International Electrotechnical Commission (IEC) 27033, 27001, 27006 and 27002, or Control Objectives for Information and Related Technology (COBIT)). The Contractor will provide an analysis for alternative methodologies compared to SP 800-53 high impact baseline identifying any gaps.

The Contractor shall deliver files to NOAA ingest using either Hypertext Transfer Protocol Secure (HTTPS) (preferred) or Secure File Transfer Protocol (SFTP). The Contractor shall use a data integrity method compliant with the NESDIS Cloud System Data Ingest and Data Distribution Services Interface Control Document (ICD) and approved by NOAA.

VII. DATA RIGHTS

Per this contract, the Contractor grants NOAA a perpetual, non-exclusive, irrevocable, worldwide license to use, reproduce, and create NOAA-derived and value-added products with the data, in any manner and for any purpose, and to authorize others to do so on its behalf (including other contractors and recipients of financial assistance awards). NOAA Level-2 data products can be shared with unlimited distribution after 24 hours.

Furthermore, the Contractor shall provide NOAA the right to distribute all data to all U.S. Government agencies and foreign government or intergovernmental entities, including National Meteorological and Hydrological Services, WMO-designated Regional Specialized Meteorological Centers, members of the Coordination Group for Meteorological Satellites, university research centers, and other entities responsible for national meteorological, space weather, emergency response, or hydrological services, immediately after receipt at NOAA, for non-commercial use but not for further distribution.

In addition to data deliveries, the Government requests specific documentation to support understanding, usage, and validation of the data. Contractors shall define the distribution rights for the documentation required. All documentation provided that is marked as 'Proprietary' or 'Confidential'—including detailed instrument specifications and satellite engineering data—shall be treated as vendor-proprietary. NOAA agrees to protect such documentation from unauthorized disclosure and will not share, publish, or distribute it to non-NOAA entities or third parties without the express written consent of the Contractor.

NOAA will ensure that any non-NOAA entity receiving restricted data, including documentation, is made aware of the identified use and distribution restrictions associated with the data sharing rights specified in this contract. If NOAA becomes aware that an entity is not adhering to those restrictions, NOAA will seek to ensure compliance, and if unsuccessful, may terminate the entity's access to the data. NOAA, however, has no additional obligations under this contract related to

data use by third parties. In no event is NOAA liable to the contractor for third party use or release of contractor data.